

Bitácora de comandos: comparación piloto de FASTQ, Submitted files y BAMs

Proyecto ENA PRJEB61942 / ERP147020 · Runs piloto ERR13828068 (AMPLICON/16S) y ERR13877920 (WGS/shotgun)

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Resumen de conclusión

Comparación	ERR13828068 (16S / AMPLICON)	ERR13877920 (WGS / shotgun)	Conclusión
BAM vs Generated FASTQ de ENA	R1 y R2: YES; mismo seq+qual MD5	R1 y R2: YES; mismo seq+qual MD5	Los BAM contienen las mismas secuencias y calidades que los FASTQ generados por ENA.
Generated FASTQ de ENA vs Submitted files de autores	Mismo número de reads, bases y longitudes; hashes coinciden al ignorar orden.	Mismo número de reads, bases, longitudes, secuencias y calidades en el mismo orden.	El contenido de secuencia/calidad es equivalente.
Diferencias observadas	Headers distintos y orden distinto entre Generated y Submitted.	Headers distintos y compresión distinta (.gz vs .bz2).	No se observó trimming, filtrado, mapeo ni cambio de calidades respecto a los FASTQ.
Uso práctico	Usar Generated FASTQ de ENA para pipeline 16S.	Usar Generated FASTQ de ENA para pipeline shotgun.	Los FASTQ generados por ENA son suficientes y más cómodos para análisis ecológico.

Nota: “Generated FASTQ” = FASTQ generados/estandarizados por ENA; “Submitted files” = archivos originales enviados por autores; “BAM files” = BAMs generados por ENA para el Run. En este piloto, los BAMs son contenedores de lecturas no alineadas.

1. Directorio de trabajo

```
cd /mnt/data/sur/users/abadia/macandog_bam_pilot
```

2. Descarga de FASTQ de ENA y Submitted files de autores

Se descendieron R1 y R2 de cada Run porque ambos experimentos son paired-end. El BAM ya estaba descendido de la bitácora anterior.

Script usado:

```
#!/bin/bash
#SBATCH --job-name=fastq_pilot
#SBATCH --partition=interactive
#SBATCH --time=10:00:00
#SBATCH --cpus-per-task=1
#SBATCH --mem=2G
#SBATCH --output=download_fastq_%j.out
#SBATCH --error=download_fastq_%j.err

set -euo pipefail

cd /mnt/data/sur/users/abadia/macandog_bam_pilot
mkdir -p fastq/generated_fastq fastq/submitted_files logs

echo "Inicio de descarga: $(date)"

echo "Descargando Generated FASTQ - ERR13828068 AMPLICON/16S"
wget -c -P fastq/generated_fastq \
  "https://ftp.sra.ebi.ac.uk/vol1/fastq/ERR138/068/ERR13828068/ERR13828068_1.fastq.gz" \
  "https://ftp.sra.ebi.ac.uk/vol1/fastq/ERR138/068/ERR13828068/ERR13828068_2.fastq.gz" \
  -o logs/ERR13828068_generated_fastq.wget.log

echo "Descargando Submitted files - ERR13828068 AMPLICON/16S"
wget -c -P fastq/submitted_files \
  "https://ftp.sra.ebi.ac.uk/vol1/run/ERR138/ERR13828068/GU85_168_L001_R1_001.fastq.gz" \
```

```

"https://ftp.sra.ebi.ac.uk/vol1/run/ERR138/ERR13828068/GU85_169_L001_R2_001.fastq.gz" \
-o logs/ERR13828068_submitted.wget.log

echo "Descargando Generated FASTQ - ERR13877920 WGS/shotgun"
wget -c -P fastq/generated_fastq \
  "https://ftp.sra.ebi.ac.uk/vol1/fastq/ERR138/020/ERR13877920/ERR13877920_1.fastq.gz" \
  "https://ftp.sra.ebi.ac.uk/vol1/fastq/ERR138/020/ERR13877920/ERR13877920_2.fastq.gz" \
  -o logs/ERR13877920_generated_fastq.wget.log

echo "Descargando Submitted files - ERR13877920 WGS/shotgun"
wget -c -P fastq/submitted_files \
  "https://ftp.sra.ebi.ac.uk/vol1/run/ERR138/ERR13877920/PLIC_SHO_LN_98_R1.fastq.bz2" \
  "https://ftp.sra.ebi.ac.uk/vol1/run/ERR138/ERR13877920/PLIC_SHO_LN_98_R2.fastq.bz2" \
  -o logs/ERR13877920_submitted.wget.log

echo "Descarga terminada: $(date)"
echo "Archivos descargados:"
find fastq -type f -exec ls -lh {} \;
```

Envío y seguimiento del job:

```

sbatch download_fastq_pilot.sbatch
squeue 237870
```

3. Verificación de archivos descargados

```

[abadia@login01 macandog_bam_pilot]$ find fastq -type f -exec ls -lh {} \;
-rw-r--r-- 1 abadia sur 12M Oct 17 2024 fastq/generated_fastq/ERR13828068_1.fastq.gz
-rw-r--r-- 1 abadia sur 14M Oct 17 2024 fastq/generated_fastq/ERR13828068_2.fastq.gz
-rw-r--r-- 1 abadia sur 2.3G Oct 23 2024 fastq/generated_fastq/ERR13877920_1.fastq.gz
-rw-r--r-- 1 abadia sur 2.3G Oct 23 2024 fastq/generated_fastq/ERR13877920_2.fastq.gz
-rw-r--r-- 1 abadia sur 15M Oct 17 2024 fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz
-rw-r--r-- 1 abadia sur 17M Oct 17 2024 fastq/submitted_files/GU85_169_L001_R2_001.fastq.gz
-rw-r--r-- 1 abadia sur 1.8G Oct 23 2024 fastq/submitted_files/PLIC_SHO_LN_98_R1.fastq.bz2
-rw-r--r-- 1 abadia sur 1.8G Oct 23 2024 fastq/submitted_files/PLIC_SHO_LN_98_R2.fastq.bz2
```

4. Comparación Generated FASTQ vs Submitted files

Objetivo: comparar conteos, bases, longitud promedio, hash de secuencias, hash de calidades y hash de IDs. Este primer hash depende del orden de las reads.

Script enviado como job:

```

#!/bin/bash
#SBATCH --job-name=compare_fq
#SBATCH --partition=interactive
#SBATCH --time=06:00:00
#SBATCH --cpus-per-task=1
#SBATCH --mem=4G
#SBATCH --output=compare_fastq_%j.out
#SBATCH --error=compare_fastq_%j.err

set -euo pipefail
cd /mnt/data/sur/users/abadia/macandog_bam_pilot
mkdir -p reports/fastq_compare

python3 << 'PY'
import gzip, bz2, hashlib
from pathlib import Path
files = [
    "fastq/generated_fastq/ERR13828068_1.fastq.gz",
    "fastq/generated_fastq/ERR13828068_2.fastq.gz",
    "fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz",
    "fastq/submitted_files/GU85_169_L001_R2_001.fastq.gz",
    "fastq/generated_fastq/ERR13877920_1.fastq.gz",
    "fastq/generated_fastq/ERR13877920_2.fastq.gz",
    "fastq/submitted_files/PLIC_SHO_LN_98_R1.fastq.bz2",
    "fastq/submitted_files/PLIC_SHO_LN_98_R2.fastq.bz2",
]
def opener(path):
```

```
if path.endswith('.gz'): return gzip.open(path, 'rt', errors='replace')
if path.endswith('.bz2'): return bz2.open(path, 'rt', errors='replace')
return open(path, 'rt', errors='replace')
def norm_id(header):
    h = header.strip()
    if h.startswith('@'): h = h[1:]
    h = h.split()[0]
    if h.endswith('/') or h.endswith('/2'): h = h[:-2]
    return h
out = Path('reports/fastq_compare/fastq_summary.tsv')
with out.open('w') as w:
    w.write('file\treads\tbases\tavg_len\tmin_len\tmax_len\tseq_md5\tqual_md5\tid_md5\tfirst_id\n')
    for f in files:
        print(f'Procesando {f}', flush=True)
        n=0; bases=0; min_len=None; max_len=0; first_id=''
        seq_md5=hashlib.md5(); qual_md5=hashlib.md5(); id_md5=hashlib.md5()
        with opener(f) as handle:
            while True:
                h=handle.readline()
                if not h: break
                seq=handle.readline(); plus=handle.readline(); qual=handle.readline()
                n+=1; seq=seq.rstrip('\n\r'); qual=qual.rstrip('\n\r'); rid=norm_id(h)
                if n==1: first_id=rid
                l=len(seq); bases+=l; min_len=1 if min_len is None else min(min_len,l); max_len=max(max_len,l)
                seq_md5.update(seq.encode()); seq_md5.update(b'\n')
                qual_md5.update(qual.encode()); qual_md5.update(b'\n')
                id_md5.update(rid.encode()); id_md5.update(b'\n')
            avg=bases/n if n else 0
        w.write(f'{f}\t{n}\t{bases}\t{avg:.2f}\t{min_len}\t{max_len}\t{seq_md5.hexdigest()}\t{qual_md5.hexdigest()}\t{id_md5.hexdigest()}\t{first_id}\n')
print('Listo. Resultado en reports/fastq_compare/fastq_summary.tsv')
PY
cat reports/fastq_compare/fastq_summary.tsv
```

Salida observada:

```
[abadia@login01 macandog_bam_pilot]$ cat compare_fastq_237900.out
Procesando fastq/generated_fastq/ERR13828068_1.fastq.gz
Procesando fastq/generated_fastq/ERR13828068_2.fastq.gz
Procesando fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz
Procesando fastq/submitted_files/GU85_169_L001_R2_001.fastq.gz
Procesando fastq/generated_fastq/ERR13877920_1.fastq.gz
Procesando fastq/generated_fastq/ERR13877920_2.fastq.gz
Procesando fastq/submitted_files/PLIC_SHO_LN_98_R1.fastq.bz2
Procesando fastq/submitted_files/PLIC_SHO_LN_98_R2.fastq.bz2
Listo. Resultado en reports/fastq_compare/fastq_summary.tsv
Resumen:
file      reads      bases      avg_len    min_len    max_len    seq_md5    qual_md5    id_md5     first_id
fastq/generated_fastq/ERR13828068_1.fastq.gz  112452     28225452   251.00     251        251        f1678b9e81cb0ef6f3a
fastq/generated_fastq/ERR13828068_2.fastq.gz  112452     28225452   251.00     251        251        d99fbfff1622f8ef56
fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz  112452     28225452   251.00     251        251        4810aee5b76499f692
M03114:37:000000000-JCMKT:1:1101:16010:1642
fastq/submitted_files/GU85_169_L001_R2_001.fastq.gz  112452     28225452   251.00     251        251        c703f7d5e7397abff4
M03114:37:000000000-JCMKT:1:1101:16010:1642
fastq/generated_fastq/ERR13877920_1.fastq.gz  31527379  4654605795  147.64     75         151        54344b089c9c946aa7
fastq/generated_fastq/ERR13877920_2.fastq.gz  31527379  4654782027  147.64     75         151        7085f6af1b7052fa3f
fastq/submitted_files/PLIC_SHO_LN_98_R1.fastq.bz2  31527379  4654605795  147.64     75         151        54344b089c9c946aa7
A00302:434:H2CV5DSX3:1:1101:10004:36479
fastq/submitted_files/PLIC_SHO_LN_98_R2.fastq.bz2  31527379  4654782027  147.64     75         151        7085f6af1b7052fa3f
A00302:434:H2CV5DSX3:1:1101:10004:36479
```

Interpretación: WGS coincidió directamente entre Generated FASTQ y Submitted files en secuencias y calidades. En 16S coincidieron conteos/bases/longitud, pero no los hashes dependientes del orden; por eso se hizo la prueba siguiente.

5. Prueba para 16S ignorando el orden

```
mkdir -p reports/fastq_compare/amplicon_order_check
for f in \
    fastq/generated_fastq/ERR13828068_1.fastq.gz \
    fastq/generated_fastq/ERR13828068_2.fastq.gz \
    fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz \
    fastq/submitted_files/GU85_169_L001_R2_001.fastq.gz
do
    base=$(basename "$f")
    echo "Procesando $base"
```

```

zcat "$f" | awk 'NR%4==2' | sort | md5sum \
> reports/fastq_compare/amplicon_order_check/${base}.sorted_sequences.md5
zcat "$f" | awk 'NR%4==2 {seq=$0} NR%4==0 {print seq "\t" $0}' | sort | md5sum \
> reports/fastq_compare/amplicon_order_check/${base}.sorted_seqqual.md5
done
for f in reports/fastq_compare/amplicon_order_check/*.md5
do
    echo -e "$(basename "$f")\t$(cut -d' ' -f1 "$f")"
done

```

Salida observada:

```

ERR13828068_1.fastq.gz.sorted_seqqual.md5      35ecc1a7bb60e78d32b3ec08a8999d70
ERR13828068_1.fastq.gz.sorted_sequences.md5    807ff0a923bb7d9733240edb075e5bb1
ERR13828068_2.fastq.gz.sorted_seqqual.md5      154df70015886569be2fcd1fbacb45f0
ERR13828068_2.fastq.gz.sorted_sequences.md5    ca91ded2cff0ab6460ea5475f110b2b7
GU85_168_L001_R1_001.fastq.gz.sorted_seqqual.md5 35ecc1a7bb60e78d32b3ec08a8999d70
GU85_168_L001_R1_001.fastq.gz.sorted_sequences.md5 807ff0a923bb7d9733240edb075e5bb1
GU85_169_L001_R2_001.fastq.gz.sorted_seqqual.md5 154df70015886569be2fcd1fbacb45f0
GU85_169_L001_R2_001.fastq.gz.sorted_sequences.md5 ca91ded2cff0ab6460ea5475f110b2b7

```

Interpretación: Generated FASTQ y Submitted files de 16S contienen las mismas secuencias y calidades; la diferencia era el orden y los headers.

6. Comparación BAM vs Generated FASTQ de ENA

Se comparó secuencia+calidad del BAM contra los Generated FASTQ de ENA, separados por R1 y R2.

```

#!/bin/bash
#SBATCH --job-name=bam_vs_fq
#SBATCH --partition=interactive
#SBATCH --time=08:00:00
#SBATCH --cpus-per-task=4
#SBATCH --mem=8G
#SBATCH --output=compare_bam_fastq_%j.out
#SBATCH --error=compare_bam_fastq_%j.err

set -euo pipefail
cd /mnt/data/sur/users/abadia/macandog_bam_pilot
mkdir -p reports/bam_vs_fastq
module load samtools/1.22.1

hash_fastq_seqqual () {
    fq="$1"
    zcat "$fq" | awk 'NR%4==2 {seq=$0} NR%4==0 {print seq "\t" $0}' | md5sum | cut -d " " -f1
}
hash_bam_r1_seqqual () {
    bam="$1"
    samtools fastq -@ 4 -n -1 /dev/stdout -2 /dev/null -0 /dev/null -s /dev/null "$bam" 2>/dev/null \
    | awk 'NR%4==2 {seq=$0} NR%4==0 {print seq "\t" $0}' | md5sum | cut -d " " -f1
}
hash_bam_r2_seqqual () {
    bam="$1"
    samtools fastq -@ 4 -n -1 /dev/null -2 /dev/stdout -0 /dev/null -s /dev/null "$bam" 2>/dev/null \
    | awk 'NR%4==2 {seq=$0} NR%4==0 {print seq "\t" $0}' | md5sum | cut -d " " -f1
}

out="reports/bam_vs_fastq/bam_vs_generated_fastq.tsv"
echo -e "run\ttype\tmate\tbam_seqqual_md5\tgenerated_fastq_seqqual_md5\tmatch" > "$out"

# 16S / AMPLICON
bam_hash=$(hash bam_r1_seqqual ERR13828068.bam); fq_hash=$(hash_fastq_seqqual fastq/generated_fastq/ERR13828068_1.fastq.gz); match="NO"; [[ "$bam_hash" == "$fq_hash" ]] && match="YES"; echo -e
"ERR13828068\tAMPLICON_16S\tR1\t$bam_hash\t$fq_hash\t$match" >> "$out"
bam_hash=$(hash bam_r2_seqqual ERR13828068.bam); fq_hash=$(hash_fastq_seqqual fastq/generated_fastq/ERR13828068_2.fastq.gz); match="NO"; [[ "$bam_hash" == "$fq_hash" ]] && match="YES"; echo -e
"ERR13828068\tAMPLICON_16S\tR2\t$bam_hash\t$fq_hash\t$match" >> "$out"

# WGS / shotgun
bam_hash=$(hash bam_r1_seqqual ERR13877920.bam); fq_hash=$(hash_fastq_seqqual fastq/generated_fastq/ERR13877920_1.fastq.gz); match="NO"; [[ "$bam_hash" == "$fq_hash" ]] && match="YES"; echo -e
"ERR13877920\tWGS_shotgun\tR1\t$bam_hash\t$fq_hash\t$match" >> "$out"
bam_hash=$(hash bam_r2_seqqual ERR13877920.bam); fq_hash=$(hash_fastq_seqqual fastq/generated_fastq/ERR13877920_2.fastq.gz); match="NO"; [[ "$bam_hash" == "$fq_hash" ]] && match="YES"; echo -e
"ERR13877920\tWGS_shotgun\tR2\t$bam_hash\t$fq_hash\t$match" >> "$out"
cat "$out"

```

```

sbatch compare_bam_vs_fastq.sbatch
queue -j 237913
cat compare_bam_fastq_237913.out
cat compare_bam_fastq_237913.err
cat reports/bam_vs_fastq/bam_vs_generated_fastq.tsv

```

run	type	mate	bam_seqqual_md5	generated_fastq_seqqual_md5	match
ERR13828068	AMPLICON_16S	R1	77858d98ca34b25ab7447df16a0c0c77	77858d98ca34b25ab7447df16a0c0c77	YES
ERR13828068	AMPLICON_16S	R2	3fca689c26a53b4baec71a31dc61599f	3fca689c26a53b4baec71a31dc61599f	YES
ERR13877920	WGS_shotgun	R1	d631bb5659b0b9e9e6ca5f8d60b5b4af	d631bb5659b0b9e9e6ca5f8d60b5b4af	YES
ERR13877920	WGS_shotgun	R2	8a8f7593b7b0c342aaab47b5da3bfe5	8a8f7593b7b0c342aaab47b5da3bfe5	YES

7. Inspección visual de una read: BAM vs FASTQ

```
module load samtools/1.22.1
samtools view ERR13828068.bam | head -n 1
zcat fastq/generated_fastq/ERR13828068_1.fastq.gz | head -n 4
zcat fastq/submitted_files/GU85_168_L001_R1_001.fastq.gz | head -n 4
```

[illegible]

Interpretación: BAM y Generated FASTQ empiezan con la misma read y la misma secuencia/calidad. El Submitted FASTQ empieza con otra read porque su orden es distinto; la prueba ordenada mostró que el contenido sí coincide.

8. Conclusión final

Conclusión resumida para reportar: En los dos Runs piloto, los BAMs, los Generated FASTQ de ENA y los Submitted files de autores contienen las mismas secuencias y calidades. Las diferencias observadas son de formato, compresión, headers/nombres de reads y, en el 16S, orden de reads. Los BAMs no contienen alineamiento ni preprocesamiento analítico extra respecto a los FASTQ.